

Project T27

Parsa Kermani Pour, Arshia sharif, Bhavdeep Arora

All of our tables are 3NF/BCNF, so we will derive BCNF for a separate example table

Given $R(A, B, C, D)$
With FD = $\begin{cases} A \rightarrow B \\ C \rightarrow D \\ D \rightarrow A \end{cases}$ $\rightarrow$ This is in 3NF because it is 2NF and has no transitive FDs

① $A^+ = \{A, B\}$ A is not superkey, so $A \rightarrow B$ violates BCNF

So we split R into R_1 and R_2

$R_1(A, C, D)$

$R_2(A, B)$

$C \rightarrow D$

$A \rightarrow B$

$D \rightarrow A$

② $C^+ = \{C, D, A\}$ C is key, so $C \rightarrow D$ does not violate BCNF
continue...

③ $D^+ = \{D, A\}$ D is not key, so $D \rightarrow A$ violates BCNF

So split R_1 into $R_{1.1}$ and $R_{1.2}$

$R_{1.1}(C, D)$

$R_{1.2}(D, A)$

$R_2(A, B)$

$C \rightarrow D$

$D \rightarrow A$

$A \rightarrow B$

④ no more violations, we are done!

$R(A, B, C, D)$ decomposed into:

$R_{1.1}(C, D)$

$R_{1.2}(D, A)$

$R_2(A, B)$

FD = $\{C \rightarrow D$

FD = $\{D \rightarrow A$

FD = $\{A \rightarrow B$